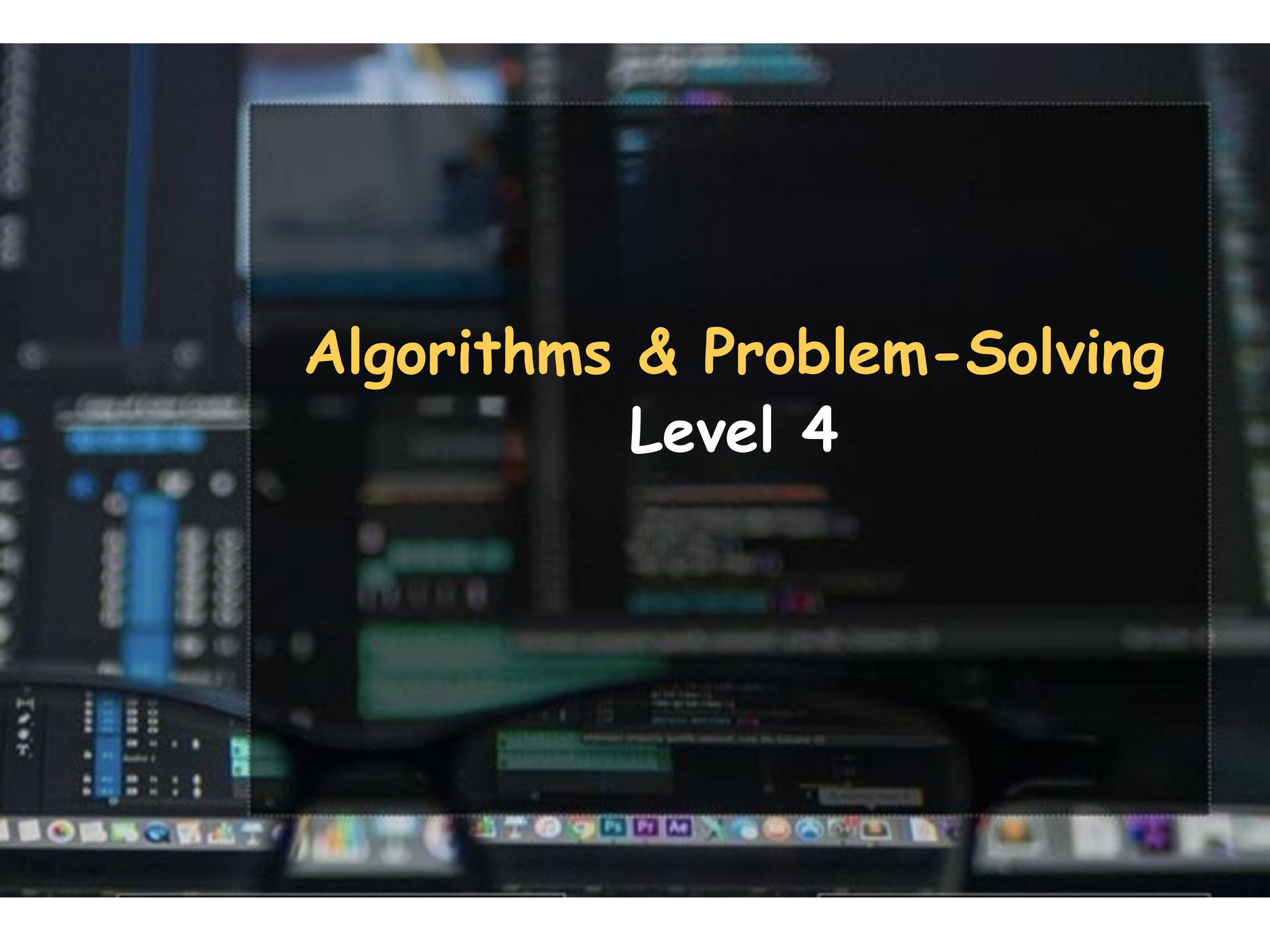
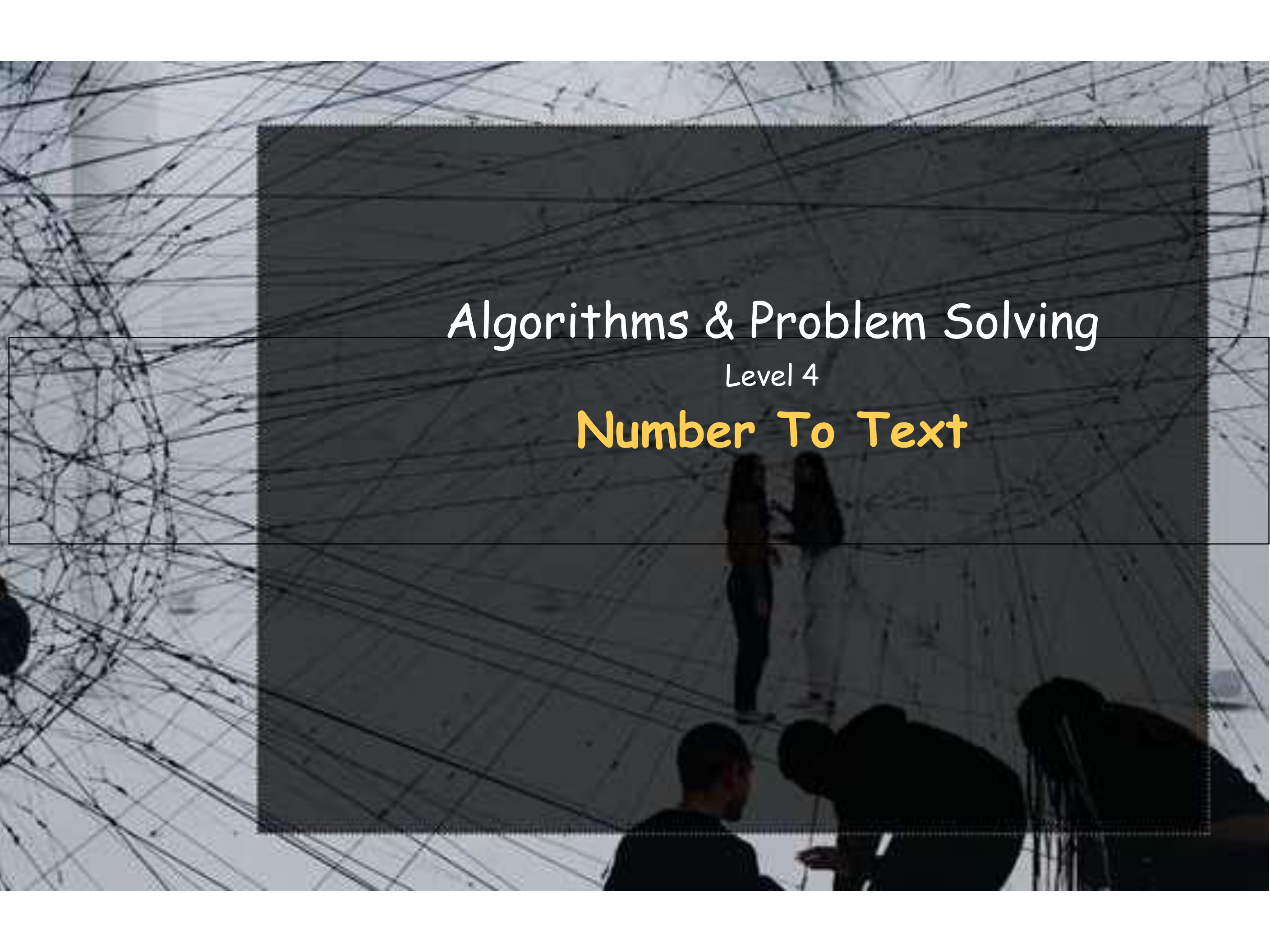




Algorithms & Problem-Solving Level 4



Algorithms & Problem Solving

Level 4

Number To Text

Definition

-  The goal is to write a C++ program that reads a number from the user and then prints it as text (English words representing the number).

Explanation:

- The program relies on functions to divide tasks:
- ``ReadNumber()`` to read the number from the user
- ``NumberToText()`` to convert the number to text based on its value
- ``main()`` to call the functions and execute the program

Procedures

① Reading the number:

-  The function ReadNumber() asks the user to enter a number and returns it as an integer (int):

```
int ReadNumber()
{
    int number;
    cout << "\nEnter a Number? ";
    cin >> number;
    return number;
}
```

② Converting the number to text:

-  The function NumberToText(int number) contains a series of conditions to check the number's value and convert it into English words.

Main cases handled:

- Units (0-9): An array `units[]` is used to convert numbers into words such as "Zero", "One", "Two", ...
- Numbers between (10-19): These are treated as a special group using a `teens[]` array, because their names do not follow the regular pattern.
- Tens (20-99): A `tens[]` array is used, and the remaining units are appended.
 - 💡 Example: 43 → Forty Three
- Hundreds (100-999): First, the hundreds part is extracted, then the remainder is handled using the modulus operator `%`.
 - 💡 Example: 584 → Five Hundred Eighty Four
- Thousands, Millions, Billions: Handled in a similar way by dividing the number according to its magnitude (thousand, million, billion) and recursively calling the same function.
 - 💡 Example: 5843 → Five Thousand Eight Hundred Forty Three



Note:

- The program relies on recursion inside the `NumberToText()` function to break the number into parts (thousands, hundreds, tens, units).
- Attention should be paid to correct word forms (for example, "Hundred" instead of "Hundreds") to improve linguistic accuracy.

Explanation

■ When the user enters 5843, the function processes it as follows:

■ $5843 \div 1000 = 5 \rightarrow \text{"Five Thousand"}$

■ Remaining 843 $\rightarrow \text{"Eight Hundred Forty Three"}$

These parts are then combined to produce the final output.

Summary

- ✓ The program reads an integer from the user and converts it into English text.
- ✓ It uses conditional logic and recursion to handle numbers from 0 up to billions.
- ✓ A practical example of using recursion and string arrays for number-to-text conversion.

Thank you 😊